Implementation of a User-Assisted Shadow Detection and Removal Algorithm Based on Thresholding and Region Growing

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***Abstract*—Shadow detection and removal is a critical task in various computer vision applications, such as object recognition and image segmentation. This paper proposes a method to detect and remove shadows from images using a combination of thresholding, region growing, and Gaussian filtering. The process starts with an initial shadow mask generated by thresholding the luminance channel of the image in the YCbCr color space. Seed points are then interactively selected for region growing, which is followed by boundary extraction and shadow region smoothing via Gaussian filtering. Experimental results demonstrate the effectiveness of the proposed approach in accurately detecting and removing shadows, improving the visual quality of images.**

Keywords: Image segmentation, Seeded Region Growing, Color image segmentation, Shadow Detection

I. Introduction

Shadow detection and removal remain significant challenges in the field of computer vision. Shadows interfere with object recognition and scene understanding by introducing substantial intensity variations, which make it difficult to distinguish between objects and their corresponding shadow regions. This complexity is further exacerbated in scenarios where shadows obscure critical details, limiting the accuracy and efficiency of image analysis systems.

The YCbCr color space provides a robust method for shadow detection by leveraging its distinct representation of luminance (Y) and chromaticity (Cb, Cr) components [1]. In this method, the luminance channel is analyzed to identify shadow regions based on pixel intensity statistics. A histogram-based dissension of the Y channel enhances the contrast, enabling precise delineation of shadow regions. Pixels with intensity values lower than one standard deviation of the image mean are initially classified as shadows, with subsequent refinement via a sliding-window iteration to ensure accurate segmentation. Morphological operations are employed to remove isolated pixels and enhance shadow boundaries, resulting in a binary shadow mask. This approach effectively identifies shadow regions while preserving the structural details of the image.

Image segmentation, as a fundamental step in addressing shadow-related issues, plays a crucial role in separating shadowed regions from the rest of the image. Among various segmentation methods, the Seeded Region Growing (SRG) algorithm stands out as a robust and intuitive approach [2]. SRG operates by selecting seed points based on specific similarity criteria and progressively expanding these regions by grouping neighboring pixels with homogeneous features. This method excels in segmenting complex image regions with smooth boundaries, making it particularly suitable for refining shadow region detection.

This paper introduces an innovative shadow detection and removal approach that combines YCbCr-based shadow detection, Seeded Region Growing (SRG), and Gaussian filtering. Initially, shadow regions are detected using the YCbCr color space to create a binary shadow mask. These regions are then refined using the SRG algorithm, which ensures spatial coherence and precise boundary delineation. Finally, Gaussian filtering is applied to smooth the segmented shadow boundaries and enhance overall image quality. By combining the strengths of YCbCr analysis and SRG for detailed boundary refinement, the proposed approach addresses the challenges of shadow-induced intensity variations and delivers superior results in object recognition and scene understanding.

II. Methodology

A. Shadow Detection

The first step in the proposed method is to detect the shadow regions within the input image. To achieve this, the image is first converted from the RGB color space to the YCbCr color space, where the luminance channel (Y channel) is extracted for further analysis. Shadows are identified as regions with lower luminance values compared to their surroundings. A thresholding technique is applied to generate the initial shadow mask, and the threshold value is calculated as:

$$\text{Threshold} = \mu_Y - \frac{\sigma_Y}{\text{Ratio}}, \tag{1}$$

where μ_Y represents the mean and σ_Y is the standard deviation of the Y channel. The `Ratio` parameter controls the sensitivity of the shadow detection process, allowing fine-tuning to adapt to varying lighting conditions. Using this threshold, the initial shadow mask is derived through the following condition:

$$\text{Shadow Mask} = Y < \text{Threshold}. \tag{2}$$

After obtaining the initial shadow mask, morphological operations are applied to refine it. Specifically, an opening operation is performed using a disk-shaped structural element. This step helps eliminate noise and small disconnected regions, ensuring a cleaner and more accurate shadow mask for subsequent processing.

As shown in Fig. 1, the shadow detection process progresses from the original image to the initial shadow mask and finally to the refined mask after the opening operation. By adjusting the `Ratio` parameter, the algorithm can effectively generate a shadow mask that accurately captures the shadow regions while minimizing false positives.

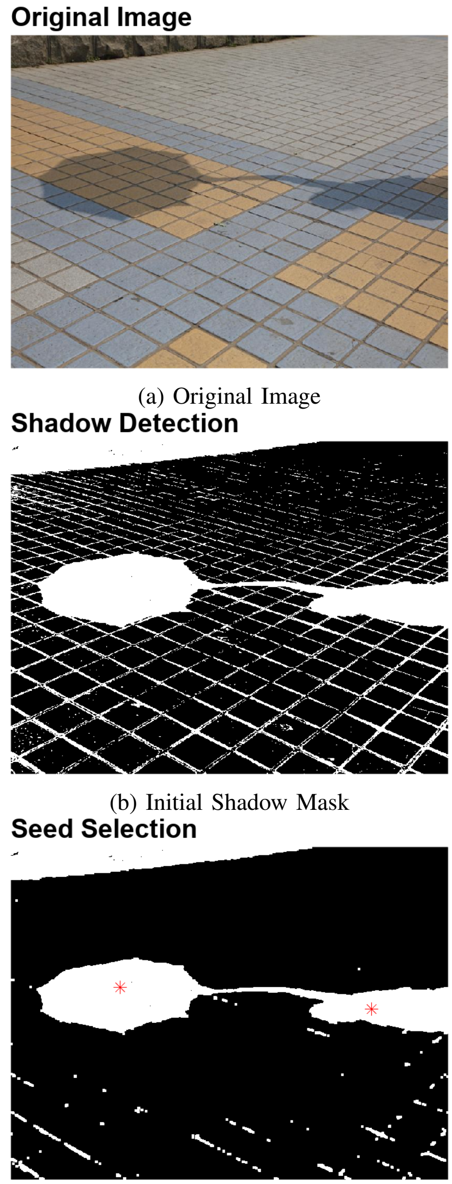
B. Region Growing

To further refine the initial shadow mask and accurately distinguish true shadow regions from non-shadow areas with lower luminance, which will also be regarded as shadow by simple thresholding, an interactive region growing technique is employed. This step ensures that only the actual shadow regions are included in the final mask, while darker non-shadow regions are excluded.

The region growing process begins with user-selected seed points, which serve as the starting locations for expansion. These seed points are chosen interactively by the user through the `ginput` function in MATLAB, ensuring that the process focuses on regions that are visually identified as shadows. The selected seed points are then validated to ensure they fall within the boundaries of the image.

Once the seed points are identified, the region growing process is performed iteratively. Using morphological dilation with a disk-shaped structuring element of radius `radius_dilate`, the region expands outward from the seed points. The expansion continues until a stable state is reached, where the grown region no longer changes between iterations. This ensures that the process fully captures the connected shadow regions without inadvertently including non-shadow areas.

The final refined shadow mask is generated by combining the regions grown from all user-selected seed points. As illustrated in Fig. 2, this step significantly enhances the accuracy of the shadow mask, providing a solid foundation for subsequent processing steps.



(a) Original Image

(b) Initial Shadow Mask

(c) Refined Shadow Mask

Fig. 1: Steps in the shadow detection process: (a) the original image, (b) the initial shadow mask generated using the thresholding technique, and (c) the refined shadow mask after applying the opening operation.

C. Boundary Extraction and Smoothing

After detecting the shadow regions, the next step is to extract the boundary between the shadow and non-shadow regions. This step is crucial for addressing the challenge of overcompensation at shadow edges, which occurs due to the gradual transition in luminance at the shadow boundary. Directly applying shadow removal to the entire shadow region can lead to overbrightening near the edges. To mitigate this, the shadow boundary is treated as a separate region for further processing.

The boundary extraction process consists of the following steps:

Boundary Region Extraction: The refined shadow mask obtained from the region growing step is subjected to morphological erosion. This operation shrinks the shadow region and isolates its inner boundary. The boundary region is extracted

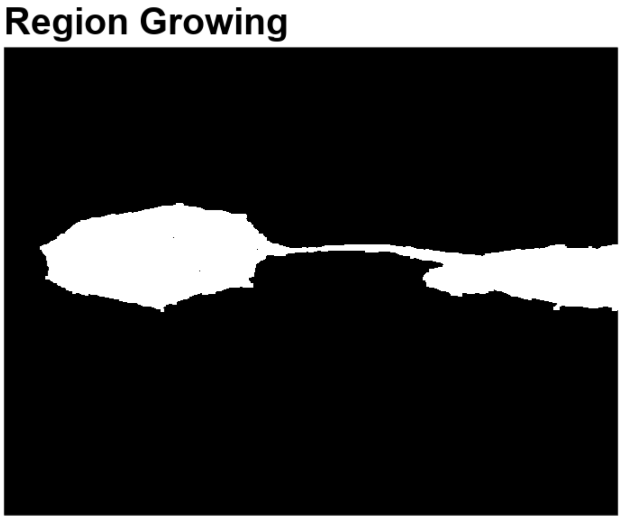


Fig. 2: The result of the region growing process, showing the refined shadow mask that accurately identifies shadow regions while excluding darker non-shadow areas.

by taking the difference between the original shadow mask and the eroded mask. This step ensures that the gradual luminance transitions at the edges of the shadow region are clearly separated from the main shadow body.

Boundary Region Expansion: The extracted boundary region is then dilated using a smaller structuring element (`radius_dilate2`). This expansion defines the region around the shadow boundary that will undergo Gaussian smoothing in subsequent steps. At this stage, the purpose is to define the boundary region for subsequent smoothing operations, ensuring a gradual and natural transition between shadow and non-shadow areas. Fig. 3 shows the extracted boundary region.

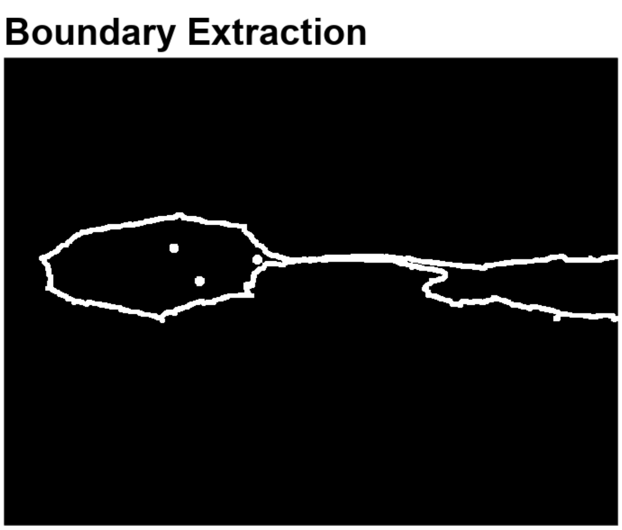


Fig. 3: Boundary extraction results

D. Shadow Removal

The final step of the proposed method is shadow removal, which is performed by adjusting the pixel values in both the shadow region and its boundary to achieve a visually seamless transition between shadowed and lit areas. This step involves three key processes:

First, for the shadow region (excluding the boundary), pixel values are adjusted using a linear transformation based on the luminance statistics of the lit and shadow regions. Specifically, the parameters α_k and γ_k are calculated for each color channel using the following equations:

$$\gamma_k = \frac{\sigma_{\text{lit}}}{\sigma_{\text{shadow}}}, \quad \alpha_k = \mu_{\text{lit}} - \gamma_k \mu_{\text{shadow}}, \tag{3}$$

where μ_{lit} and μ_{shadow} are the mean pixel values of the lit and shadow regions, respectively, and σ_{lit} and σ_{shadow} represent their standard deviations. The pixel values in the shadow region are then adjusted using the transformation:

$$I_{\text{corrected}} = \alpha_k + \gamma_k I_{\text{shadow}}. \tag{4}$$

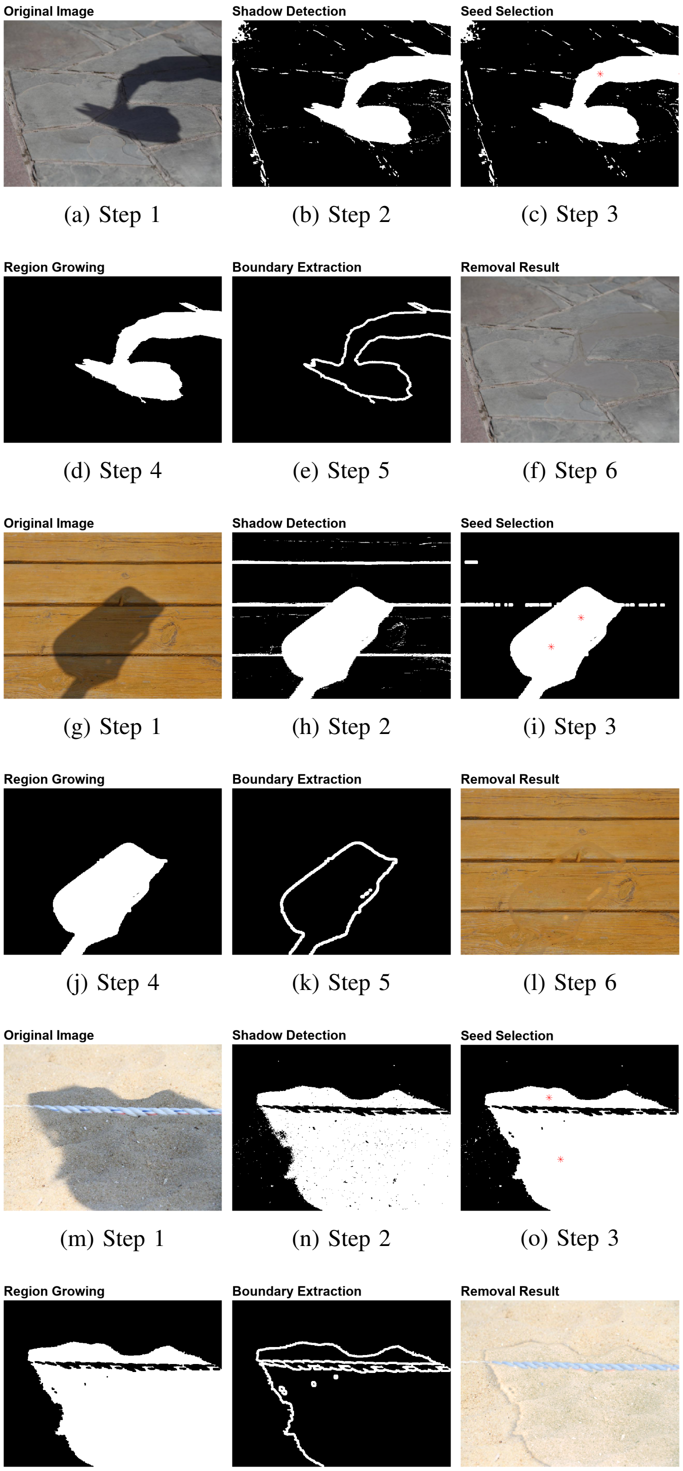
Second, for the boundary region, the compensation parameters α_1 and γ_1 are recalculated based on the statistics of the boundary region and the lit area. The same linear transformation is applied to the boundary region using the updated parameters. This ensures a more accurate correction that accommodates the gradual transition in luminance at the boundary.

Finally, Gaussian filtering is applied to the smooth boundary region identified in the previous step. This operation ensures a visually seamless and gradual transition from the shadow region to the lit area by softening any remaining sharp edges. Fig. 4 shows the results of the shadow removal process.

Fig. 4: Final shadow-removed result.

III. Experimental Results and Evaluation

The proposed shadow detection and removal method was evaluated on several images under varying lighting conditions. The step-by-step results, as shown in Figure 5, illustrate the effectiveness of the method in detecting and removing shadows



(a) Step 1 (b) Step 2 (c) Step 3

(d) Step 4 (e) Step 5 (f) Step 6

(g) Step 1 (h) Step 2 (i) Step 3

(j) Step 4 (k) Step 5 (l) Step 6

(m) Step 1 (n) Step 2 (o) Step 3

(p) Step 4 (q) Step 5 (r) Step 6

Fig. 5: Shadow detection and removal process for different images.

while preserving the underlying details. The YCbCr color space was used for shadow detection, where the luminance channel was thresholded to generate the initial shadow mask. Through interactive seed point selection, the region growing process refined the shadow boundaries, and Gaussian filtering was applied to smooth the boundary region, ensuring a more natural-looking transition from shadow to non-shadow areas. The results demonstrate that the combination of thresholding, region growing, and linear transformation effectively removes shadows, maintains texture continuity, and adapts to diverse scenes under different lighting conditions.

However, the method also has certain limitations. The boundary correction, while improved by Gaussian smoothing, is not always perfect, especially for shadows with highly irregular or sharp edges. In some cases, artifacts may remain visible along the shadow boundaries. Additionally, when the illuminated region and the shadowed region belong to different objects or exhibit distinct chromatic properties, the restored shadow region tends to inherit a color bias toward the illuminated area, leading to unnatural color transitions. Furthermore, the algorithm's performance degrades for images with overly complex shadow patterns, intricate textures, or multiple light sources, as these scenarios challenge the assumptions of uniform lighting and chromatic consistency. Despite these limitations, the method achieves satisfactory results for most practical scenarios and provides a solid foundation for further development.

IV. Conclusion and Discussion

In this paper, we proposed a shadow detection and removal framework combining thresholding, region growing, and linear transformation, with additional boundary refinement using Gaussian smoothing. The method effectively detects shadows and removes them from images, improving the overall visual quality and preserving the underlying texture. The step-by-step approach demonstrated robustness in handling varying lighting conditions and ensured smooth transitions between shadowed and illuminated regions. The results validate the effectiveness of the method in most practical scenarios, particularly for relatively simple images with well-defined shadows.

Despite its effectiveness, the proposed method has limitations that warrant further improvement. The boundary processing, although enhanced with Gaussian smoothing, occasionally leaves artifacts along sharp or irregular shadow boundaries. Additionally, the method struggles with cases where the illuminated and shadowed regions have significantly different chromatic properties, causing color biases in the restored shadow areas. For highly complex scenes with intricate shadow patterns or multiple light sources, the detection and removal performance declines noticeably, reflecting the challenges of balancing accuracy and generalization in such scenarios.

Future work will focus on addressing these challenges. To improve boundary refinement, more sophisticated edge-preserving techniques, such as guided filtering or adaptive smoothing, can be explored. Incorporating contextual information, such as object segmentation or material properties, could help mitigate color biases in shadow correction and enhance the algorithm's applicability to scenes with heterogeneous lighting. Furthermore, integrating deep learning approaches, such as convolutional neural networks (CNNs) or generative adversarial networks (GANs), may improve both the detection accuracy and the robustness of shadow removal in complex and dynamic environments. By addressing these aspects, the proposed framework can be extended to more diverse applications, including real-time video processing, autonomous navigation, and advanced image editing tasks.

In summary, this work provides a foundation for shadow detection and removal with promising results, while highlighting areas for improvement and further research. The proposed method represents a step forward in enhancing image quality under challenging lighting conditions and serves as a basis for future advancements in this field.

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3) Lanxuan Zhang: Related-work collecting, Report Writing, Oral Presentation(30%)

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